

Intro to IP Addressing (Part 1)

- What is an IP Address?
- What's the purpose of an IP Address?
- IPv4 & IPv6
- IP Address basics – bits & bytes
- Decimal & Binary

What is an IP Address?

- A number which identifies a device on a network
- Configured in software

What's the Purpose of an IP Address?

- Enables hosts to communicate across networks
- If you want to use TCP/IP then you must have an IP Address

IPv4 & IPv6

- IPv4
 - “internet protocol”
 - approximately 4.3 billion IP addresses
 - Nearly exhausted
- IPv6
 - Newer version of IP to succeed version 4
 - How many? 2^{128}

Bits & Bytes

- Bit is either a 1 or a 0 (zero)
- Byte = 8 bits
- IP Address is made up of 32 bits

Decimal & Binary

- Can write an IP address in two ways
- Binary

0000 1010 0000 1010 0000 1010 0000 0001

- Decimal

10.10.10.1

Summary

- Study IP Addressing every day!
- IP addresses identify devices & enable devices to communicate across networks
- IPv4 & IPv6
- 8 bits make up 1 byte (“octet”)
- IP address has 4 octets
- Use dotted-decimal notation
- Octet value range: 0 - 255

Intro to IP Addressing – Part 2

- Groups of IP addresses
- Two portions of the IP address

Summary

- IP addresses are grouped
- Benefit of grouping: more efficient routing
- Portions of an IP address
 - Network = common to all group members
 - Host = unique to each group member

	1 st Octet Range	Network Numbers
Class A	1 to 126	1.0.0.0 to 126.0.0.0
Class B	128 to 191	128.0.0.0 to 191.255.0.0
Class C	192 to 223	192.0.0.0 to 223.255.255.0

Summary

- 3 Classes of Network
- Each class is identified the by a range in the first octet
- Each class has a unique number of networks and hosts available

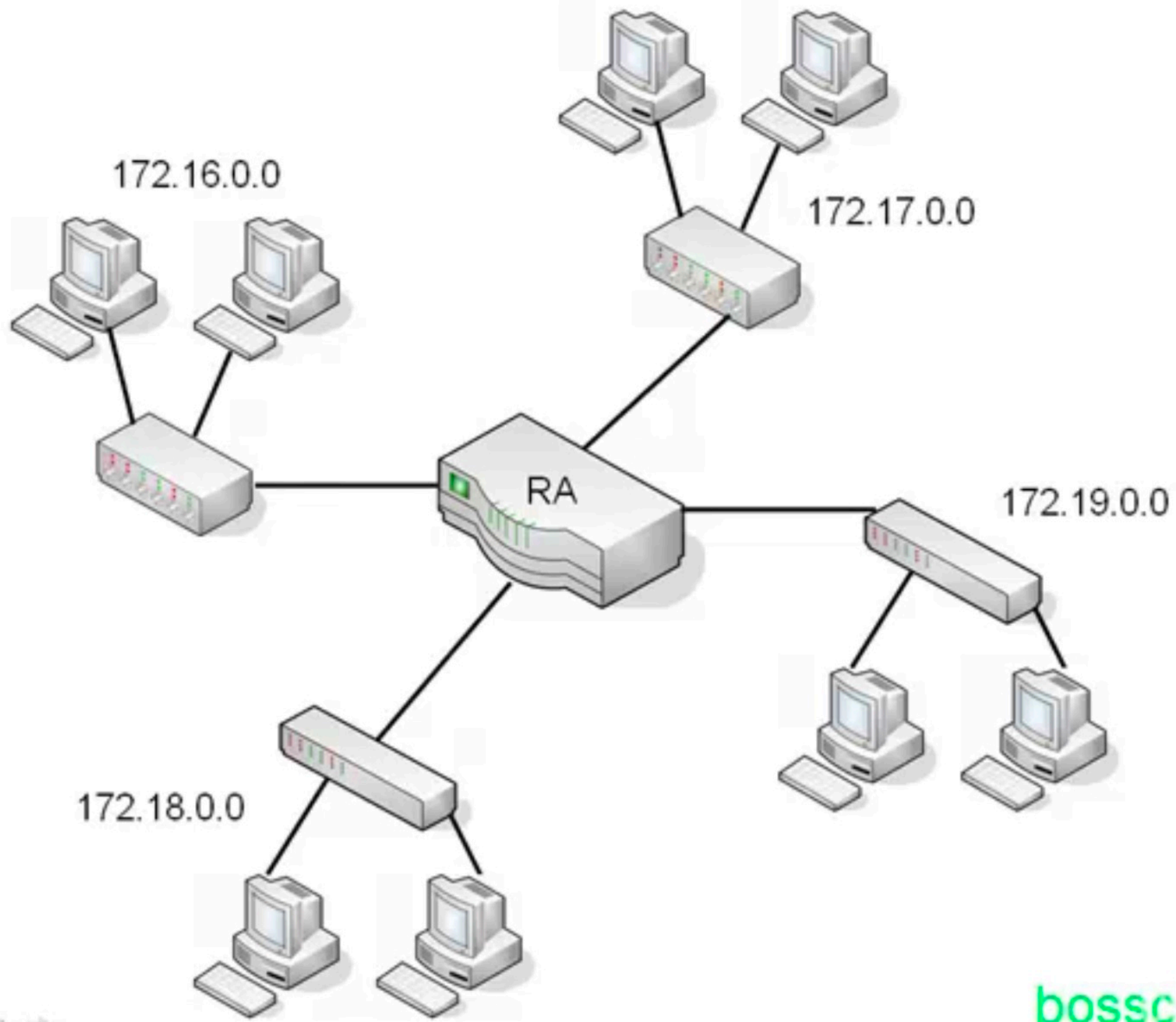
Intro to IP Subnetting

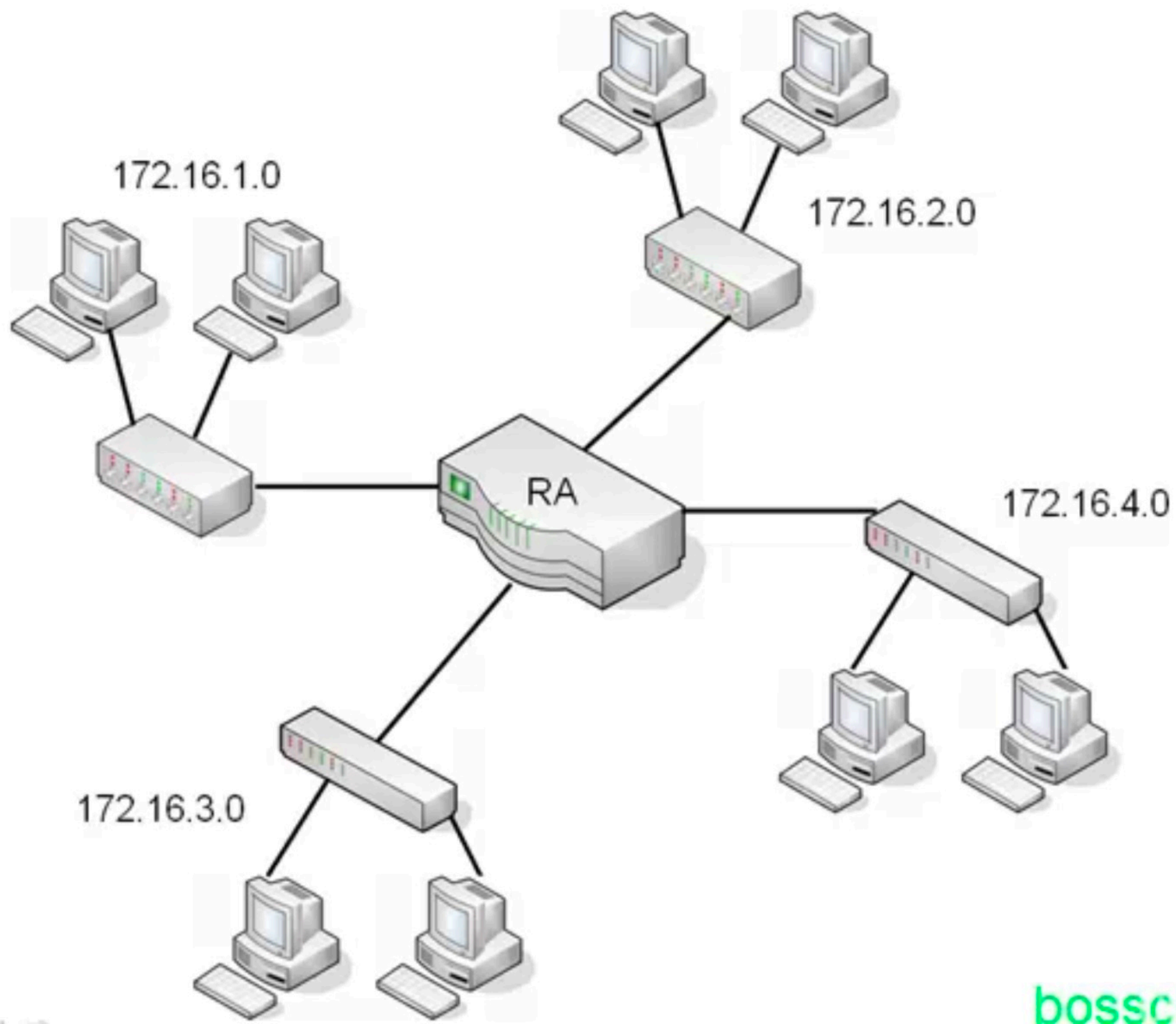


Class A
1 - 126

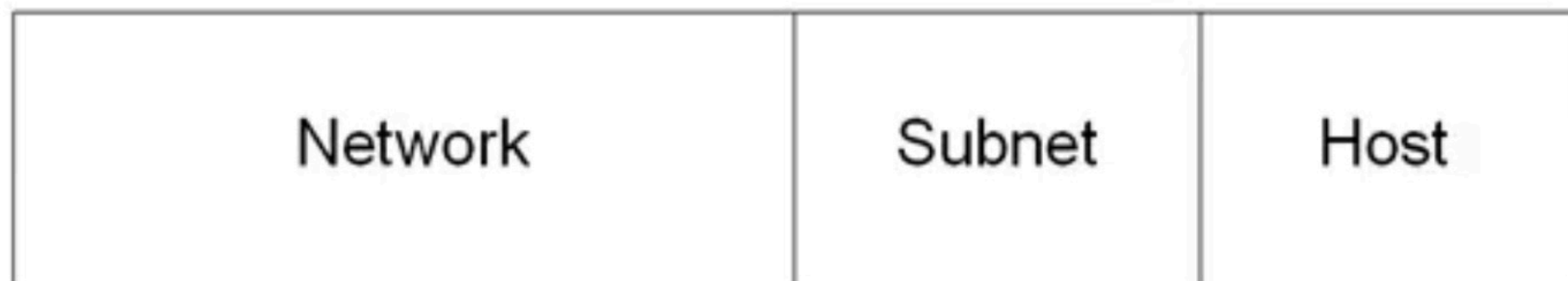
Class B
128 - 191

Class C
192 - 223





- Subnet portion of the IP address
 - Taken from the host portion
 - Network portion never changes



Summary

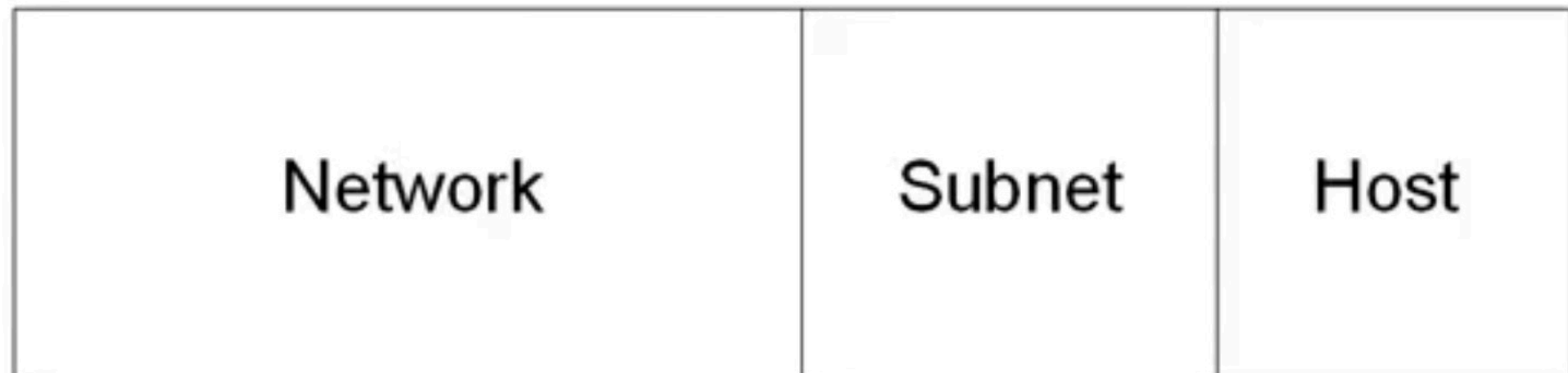
- Subnetting enables more efficient use of the network classes
- Take bits from the host portion to create the subnet portion
- Router considers the network and subnet portions when routing

Classful Addressing

- Network portion follows the rules of the Network Classes
- Class A = Network portion is the first Octet
- Class B = Network portion is Octets 1 & 2
- Class C = Network portion is Octets 1 - 3

Classless Addressing

- IP Addresses have only two parts
 - Combined Network/Subnet Portion
 - The Host portion



Intro to the Subnet Mask

- 32 bits long
 - 255.255.0.0
 - 11111111.11111111.00000000.00000000
- Accompanies the IP Address
- Network portion = 1s
- Host portion = 0s

Network Classes & Default Subnet Masks

- Network portion of Class A, B, and C addresses never changes (no subnetting)

- Class A

- IP Network ID = 10.0.0.0



- Binary Subnet Mask =

11111111.00000000.00000000.00000000

- Decimal Subnet Mask = 255.0.0.0

Network Classes & Default Subnet Masks

- Class B

- IP Network ID = 172.16.0.0



- Binary Subnet Mask =

11111111.11111111.00000000.00000000

- Decimal Subnet Mask = 255.255.0.0

Network Classes & Default Subnet Masks

- Class C

- IP Network ID = 192.168.1.0



- Binary Subnet Mask =

11111111.11111111.11111111.00000000

- Decimal Subnet Mask = 255.255.255.0

Subnet Masks When Subnetting is Used

- You don't always steal an entire octet
 - Sometimes you only take a portion of the host portion
- Subnet Number = 192.168.1.0
 - Subnet Mask = 255.255.255.240

11111111.11111111.11111111.11110000

CIDR

(Classless Inter-Domain Routing)

10.0.0.0

255.0.0.0

How many hosts are in this subnet?

- Subnet mask is comprised of 1s and 0s
- 1s are always consecutive
- *subnet / # of 1 bits in subnet mask*
- 10.0.0.0/8
 - Subnet mask equals

11111111.00000000.00000000.00000000

Network Classes

- Class A
 - 10.0.0.0/8
- Class B
 - 172.16.0.0/16
- Class C
 - 192.168.0.0/24

Summary

- Prefix notation
- Number of 1s bits in the subnet mask
- Know network & host portions
- Know the subnet mask
- Know size of the subnet (i.e. # of hosts)

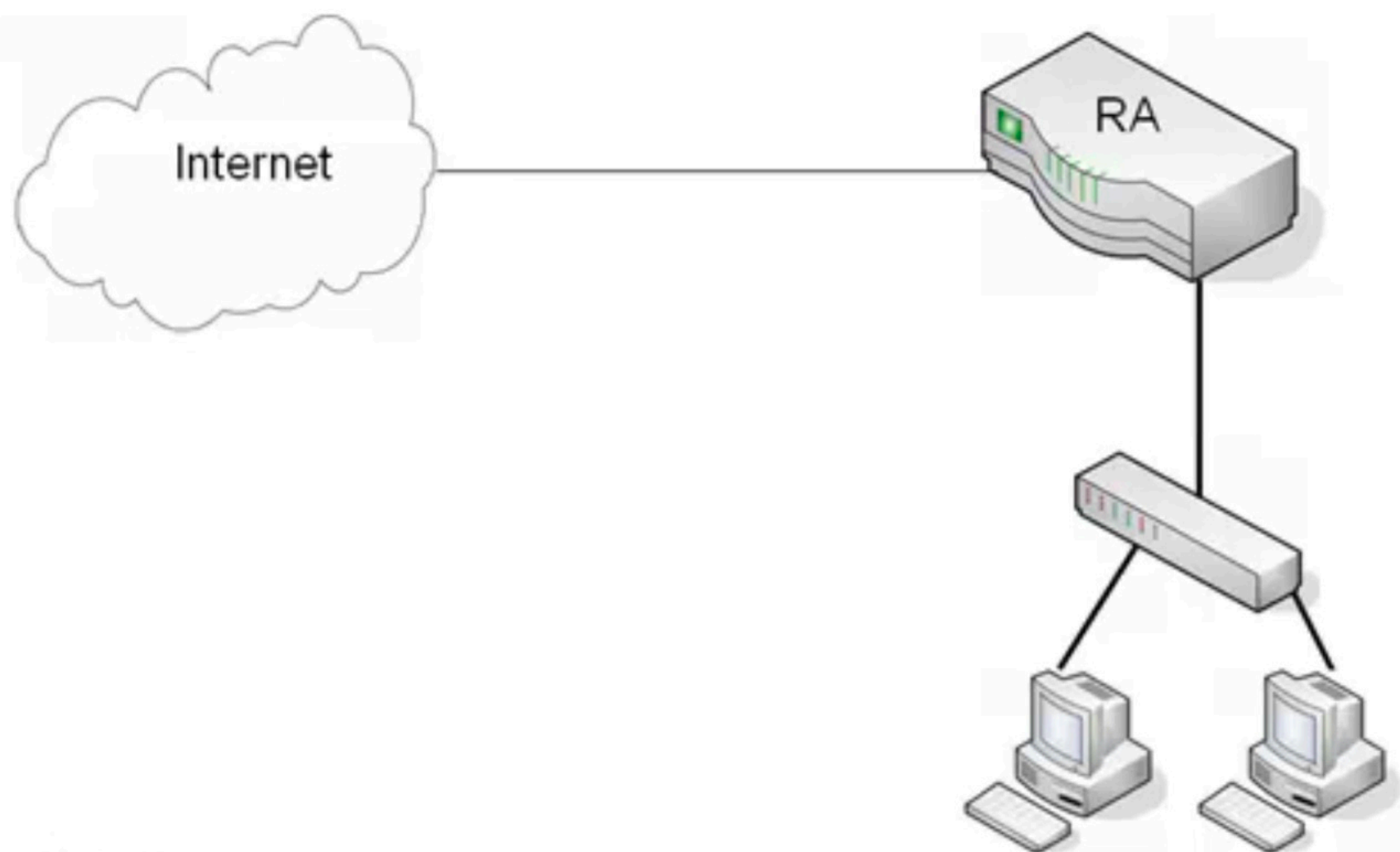
Public & Private IP Addresses

- Public IPs
 - Assigned by ICANN
 - Routable over the Internet
- Private IPs
 - Not assigned by ICANN
 - Not routable over the Internet
 - Used on private networks
 - Often referred to as RFC1918 IP space

Private IP Networks

- Class A 10.0.0.0
 - 10.0.0.0 through 10.255.255.255
- Class B 172.16.0.0
 - 172.16.0.0 through 172.16.31.255
- Class C 192.168.0.0
 - 192.168.0.0 through 192.168.255.255

IP Addresses & NAT



Summary

- Memorize the RFC1918 Networks
- Private IPs = Not routable
- Conserve Public IPs
- NAT

Decimal to Binary Conversion

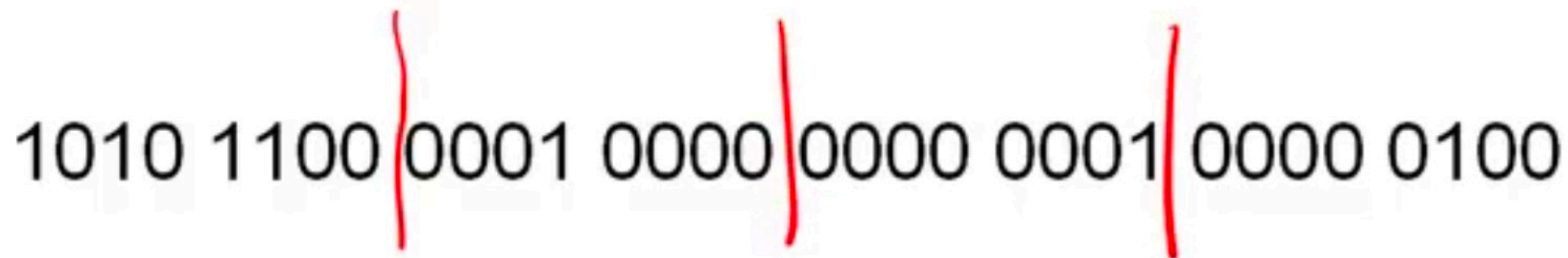
120 to Binary

128 64 32 16 8 4 2 1

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Binary to Decimal Conversion

1010 1100 0001 0000 0000 0001 0000 0100



8 bits = 1 octet

4 octets in an IP address

Summary

- Base Two: 0 or 1
- Base Ten: 0 1 2 3 4 5 6 7 8 9
- Values 2^x
 - 128 64 32 16 8 4 2 1
- On or Off

VLSM & FLSM



- Variable-Length Subnet Masking

Subnets can
be different
sizes

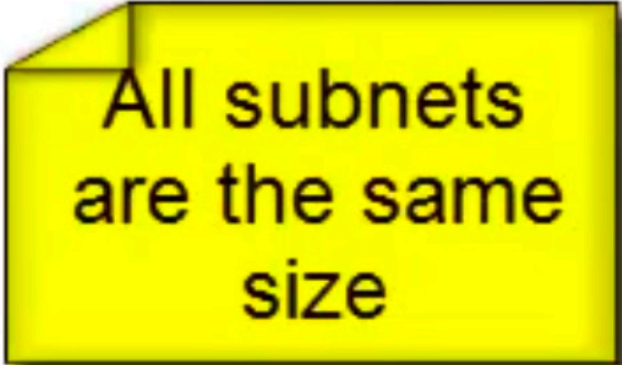
Different
subnet
masks

VLSM & FLSM

- Variable-Length Subnet Masking
- Fixed-Length Subnet Masking

A yellow sticky note with a folded top-left corner, containing the text "static-length".

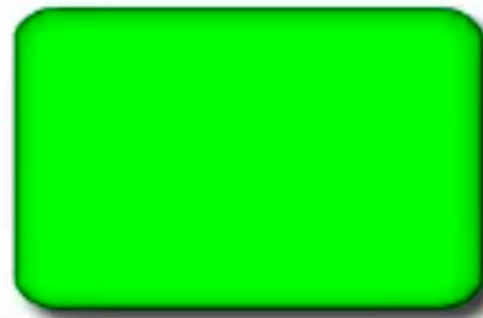
"static-length"

A yellow sticky note with a folded top-left corner, containing the text "All subnets are the same size".

All subnets
are the same
size

VLSM & FLSM

- Variable-Length Subnet Masking



- Fixed-Length Subnet Masking

"static-length"

All subnets
are the same
size

Same
subnet mask
used for all

VLSM & FLSM

- Variable-Length Subnet Masking

RIPv2

OSPF

EIGRP

- Fixed-Length Subnet Masking

RIPv1

FLSM

- 192.168.1.0/24
 - 192.168.1.0/26
 - 192.168.1.64/26
 - 192.168.1.128/26
 - 192.168.1.192/26

VLSM

- 192.168.1.0/24
 - 192.168.1.0/25
 - 192.168.1.128/26
 - 192.168.1.192/27
 - 192.168.1.224/28
 - 192.168.1.240/28

→ 192.168.1.0 255.255.255.0

8

8

8

128

① 64
32

①

• 11111111.11111111.11111111.00000000

• 4th Octet becomes 10000000 (= 128)

• New Subnet mask = 255.255.255.128 (/25)

• How many subnets? $2^1 = 2$

• How many hosts per subnet? $2^7 = 128$

• Subnet Range 0 – 127

• Subnets 192.168.1.0 & 192.168.1.128

• Valid hosts? $2^7 - 2 = 126$

192.168.1.0/25

Subnet Mask:
255.255.255.128

Range: 0 - 127

Valid Hosts:
1 - 126

Broadcast IP:
192.168.1.127

192.168.1.128/25

Subnet Mask:
255.255.255.128

Range: 128 - 255

Valid Hosts:
129 - 254

Broadcast IP:
192.168.1.255

- Valid hosts? $2^7 - 2 = 126$

192.168.1.0 255.255.255.0

- 11111111.11111111.11111111.00000000
- 4th Octet becomes 11111000 (= 248)
- New Subnet Mask = 255.255.255.248
(/29)
- How many subnets? $2^5 = 32$
- How many hosts per subnet? $2^3 = 8$
- Subnet range: 0 – 7
- Subnets: 192.168.1.0, .8, .16, .24
- Valid hosts? $2^3 - 2 = 6$

8 10.0.0.0 255.0.0.0

• 11111111.00000000.00000000.00000000

• 2nd Octet becomes: 10000000 (= 128)

• New Subnet Mask: 255.128.0.0 (/9)

• How many subnets? $2^1 = 2$

• How many hosts per subnet? $2^{23} =$

8,388,608

0 - 127

10.128.0.0

$$\begin{array}{r} 256 \\ - 128 \\ \hline 128 \end{array}$$

• Subnet Range 10.0.0.0 – 10.127.255.255

• Subnets: 10.0.0.0 & 10.128.0.0

• Valid hosts? $2^{23} - 2 = 8,388,606$

172.16.0.0 255.255.0.0

- 11111111. 11111111.00000000.00000000
- 3rd Octet becomes: 11100000 (= 224)
- New Subnet Mask: 255.255.224.0 (/19)
- How many subnets? $2^3 = 8$
- How many hosts per subnet? $2^{13} = 8192$
- Subnet Range 172.16.0.0 – 172.16.31.255
- Subnets: 172.16.0.0, 172.16.32.64, etc.
- Valid hosts? $2^{13} - 2 = 8190$

The Zero & Broadcast Subnets

- Zero Subnet = the first subnet created from a Classful network
- Broadcast Subnet = the last subnet created from a Classful network

192.168.1.0/24

New Subnet Mask:

255.255.255.192

2

64/26

192.168.1.0/26

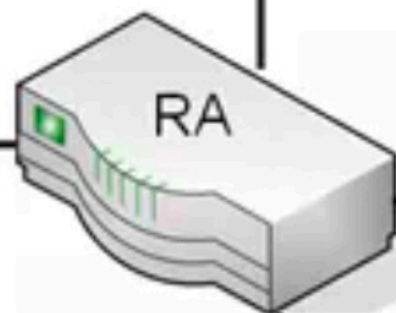
256
- 192

64

128/26

1100 0000

62



50
192/26

192.168.1.0 -
192.168.1.64 -
192.168.1.128 -
192.168.1.192 -

bosschi tv

Summary

- Know the formulas!
- Number of subnets: $2^{\text{subnet bits}}$
- Number of hosts: $2^{\text{host bits}}$
- Subnet Numbers: 256 – “new octet #”
- Review Structure of IP address & Subnet Mask
- Review Binary 2^y

Find the Number of Subnets & the Number of Valid Hosts per Subnet

- Subnets

- $2^{\text{\# of subnet bits}}$

- Valid Hosts

- $2^{\text{\# of host bits}} - 2$

network
Broadcast

192.168.1.0

255.255.255.240 (/28)

11111111.11111111.11111111.11110000
Network

- Is this a Class A, B, or C network?

- Subnets = $2^4 = (2 \times 2 \times 2 \times 2) = 16$

- Hosts = $2^4 = 16 - 2 = 14$

192.168.2.33 255.255.255.248

8+8

16+8

24+8

32+8

40-1=39

192.168.2.33

255.255.255.248

192.168.2.39

256
-248

8

Broadcast Address = 192.168.2.39

Summary

- 2 rules
 - 255 in the subnet mask = write down IP address octet value
 - 0 in the subnet mask = write down 255
- Funny octets != 255 or 0
- Use the formula: $256 - \text{funny octet} = \text{multiplier}$
 - Find number greater than IP address funny octet but as close to it as possible, then subtract 1.

How to Find All Subnet Numbers

- Focus on the subnet mask octet that does **not** equal 255 or 0.
- 192.168.2.0/24
 - Find all subnets created when this subnet mask is applied: 255.255.255.224

172.16.0.0 255.255.0.0

- Subnet Mask: 255.255.255.224

- Formula: $256 - \underline{224} = \underline{32}$

172.16.2.0/27

255

- 172.16.0.0/27 ←
- 172.16.0.32/27
- 172.16.0.64/27
- 172.16.0.96/27
- 172.16.0.128/27
- 172.16.0.160/27
- 172.16.0.192/27
- 172.16.0.224/27
- 172.16.0.256 << not valid!

- 172.16.1.0/27
- 172.16.1.32/27 —
- 172.16.1.64/27 —
- 172.16.1.96/27 —
- 172.16.1.128/27
- 172.16.1.160/27
- 172.16.1.192/27
- 172.16.1.224/27
- 172.16.1.256 << not valid!

Subnetting Used

$$16 + 16 + 16$$

$$32$$

$$48$$

$$+ 16$$

$$\boxed{64}$$

192.168.20.73

255.255.255.240

192.168.20.64

$$\begin{array}{r} 256 \\ - 240 \\ \hline 16 \end{array}$$

Subnet Number = 192.168.20.64

Summary

- For each 255 octet in the subnet mask, copy the IP Address octet value
- For each 0 octet in the subnet mask, write a 0 for the subnet number octet
- If an octet $\neq$ 255 or 0, use the formula.

172.16.20.19 255.255.240.0

$$\begin{aligned} 16 + 16 \\ = 32 - 1 \\ = 31 \end{aligned}$$

172.16.20.19
255.255.240.0

172.16.31.255

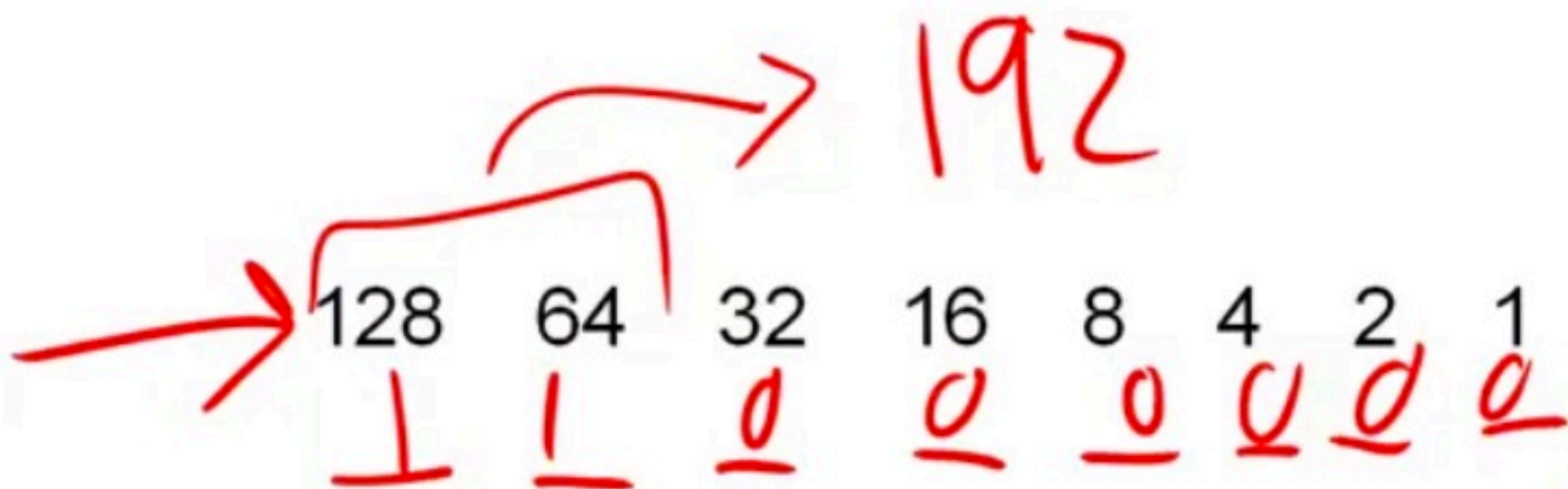
$$\begin{aligned} 256 \\ - 240 \\ \hline 16 \end{aligned}$$

Broadcast Address = 172.16.31.255

Summary

- 2 rules
 - 255 in the subnet mask = write down IP address octet value
 - 0 in the subnet mask = write down 255
- Funny octets $\neq$ 255 or 0
- Use the formula: $256 - \text{funny octet} = \text{multiplier}$
 - Find number greater than IP address funny octet but as close to it as possible, then subtract 1.

Converting Dotted-Decimal to Prefix Notation



255.255.255.192 = /26

Subnet Masks – One Size Fits All?

Design Requirements:

- 65 Subnets
- 150 Hosts per subnet
- Use 172.16.0.0/16

172.16.0.0/16

11111111.11111111.00000000.00000000

14

65 Subnets

⑦

- Determine minimum number of subnet bits required

$$2^2 = 4, 2^6 = 64, 2^7 = 128$$

150 Hosts

- Determine minimum number of host bits required

$$2^8 = 256 \text{ Host bits}$$

11111111.11111111.SSSSSSSS⁷X⁸HHHHHHHH
↓
↑

7 Subnets, 9 Host bits

- 255.255.254.0

~~512~~ 512 Hosts

- 255.255.255.0

8 subnet, 8 Host
256

Summary

- Sometimes more than one subnet mask will meet the requirement
- Examine all options

Intro to IPv6

ICANN

Internet Corporation for
Assigned Network Numbers

Intro to IPv6

IPv4

2^{32}

IPv6

2^{128}

Intro to IPv6

IPv4 & IPv6 are
not interoperable

New IP Headers

⊙ Smaller = more efficient

- Faster look-ups in route tables
- Can modify with extensions

Address Size

➔ IPv4 = 32 bits

- IPv6 = 128 bits

No More NAT & PAT



NAT = developed to help extend the availability of IPv4 addresses

- IPv6 = so many addresses we just don't need these tools

No Broadcast Packets

- No more broadcast storms!
- IPv6 uses multicast packets instead

IPSec

- Required on IPv6 hosts
- More integrated into IPv6 design

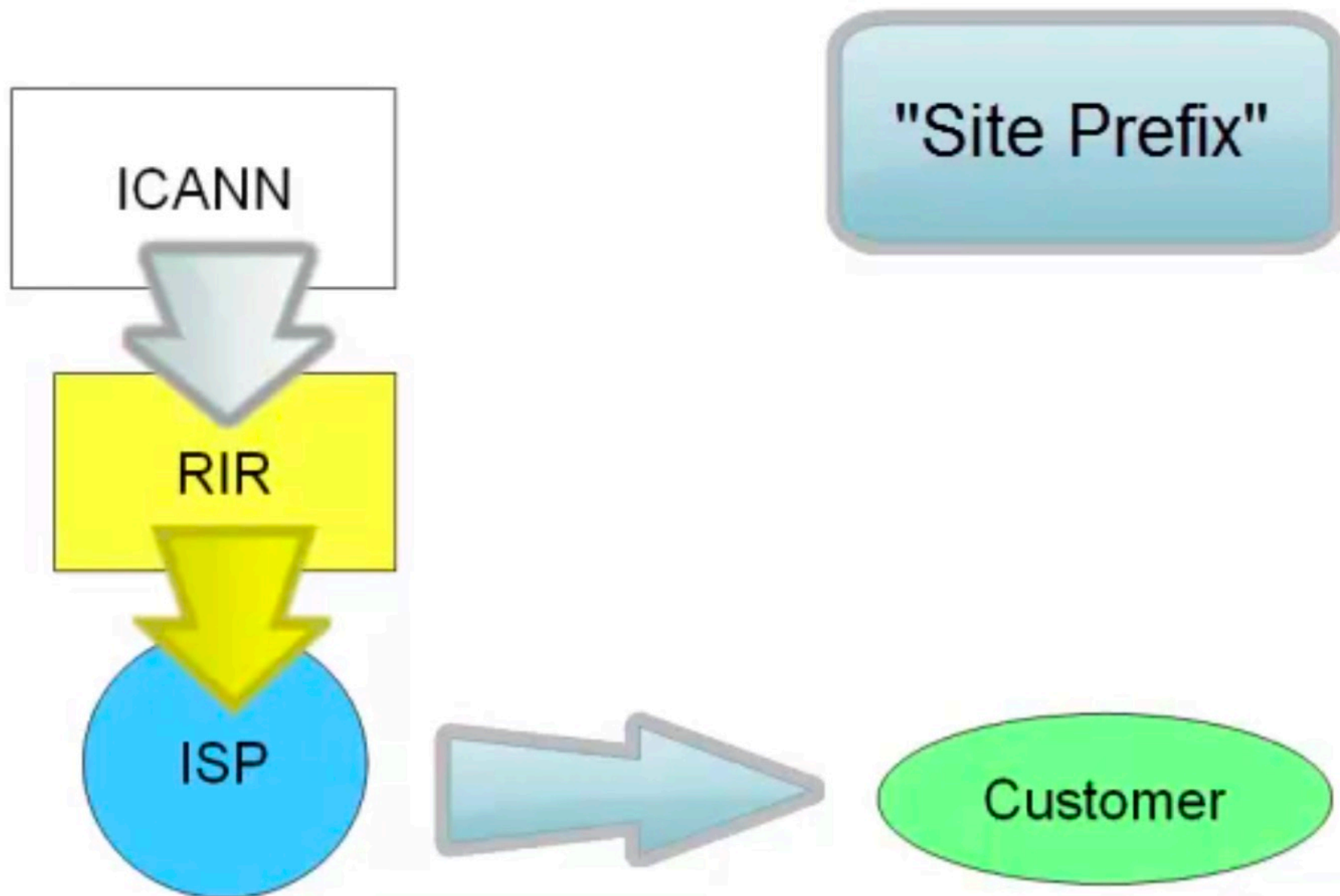
Address Aggregation

- Better aggregation with IPv6
- We get more efficient route tables

No More Fragmentation

- IPv6 determines path MTU before sending packets
- Packets that exceed MTU are just dropped

Address Assignment



Summary

- New Headers
- Address Size
- No NAT/PAT
- No Broadcast
- IPSec
- Aggregation
- Fragmentation
- Address Allocations

